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Environmental life cycle assessment of textile bio-recycling – valorizing cotton-polyester textile waste to pet fiber and glucose syrup

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ABSTRACT

Emerging textile bio-recycling approaches can address the environmental challenges associated with the end-of-life of clothing. However, it is imperative to assess the environmental impacts associated with these technologies, systematically, to ensure they are environmentally sustainable. The objective of this study is to evaluate the environmental implications of a novel bio-recycling method that recovers polyester (PET) fibres and glucose, from 50/50 cotton/PET blend of fabric waste from H&M, using Life Cycle Assessment (LCA). The chosen functional unit is one kg of recovered PET fibre. Life Cycle Impact Assessment (LCIA) was conducted in terms of ReCiPe, both midpoint and endpoint indicators, and Cumulative Energy Demand (CED) impact categories. LCA results of the gate-to-cradle analysis indicate pre-treatment as the most dominant process, followed by melt-spinning and then enzymatic hydrolysis. Sensitivity analysis with Global Warming Potential (GWP) indicator shows that pre-treatment step influences the results and contributes to uncertainty. Pre-treatment (207 MJ) is also the most energy-intensive step, followed by melt-spinning (98.5 MJ) and enzymatic hydrolysis (44.8 MJ). We also describe linkages between addition of PET bottles in the melt-spinning step and environmental impacts. Environmental impacts on all three endpoints increased as the percentage of waste PET bottle chips added was decreased. LCA of the textile bio-recycling method suggests that environmental impacts can be further reduced, provided the involved unit processes are made more energy-efficient, and the fibre quality of the recovered polyester is improved such that it can be directly used for garment production without the need to add waste PET bottle chips.

1. Introduction

Increasing textile waste is a global problem, and many studies have indicated that most of the textile waste goes to landfills (Nørup et al., 2019). The textile manufacturing sector has emerged as one of the largest contributors to greenhouse gas emissions globally. Each year textile manufacturing processes produce more than 1.2 billion tonnes of carbon emissions (Ellen MacArthur foundation, 2017). Global per capita textile consumption rates are expected to only increase in the future with fast changing-trends, increasing world population, and rising purchasing power, particularly in developing countries (Terinte et al., 2014). This consumption will drive the waste generation and consequently burden the landfills. For instance, in the US, textiles account for 4.5% of the total Municipal Solid Waste (MSW) in landfills (World Trade Statistical Review, 2018). This problem is at its acutest at the urban scale, especially in cities with limited landfilling space. In Hong Kong, 343 tonnes of textile waste is dumped daily into landfills, and unfortunately, neither cotton nor human-made fibre-based textiles were recycled as of 2017 (Environmental Protection Department 2017). Geopolitical uncertainty surrounding the trade of waste further emphasizes the need to explore opportunities to valorize textile waste and enhance the recycling capacity locally. China's foreign waste ban that includes restrictions on discarded textile materials serves as an impetus to expand local solid waste management and recycling industries (Qu et al., 2019).

Due to the challenges mentioned earlier, research efforts on recycling techniques for textile waste streams have gained attention. Potential environmental benefits of conventional textile recycling approaches based on mechanical, chemical, and thermal techniques have been assessed using Life Cycle Assessment (LCA) in the literature (Dahlbo et al., 2017; Muthu et al., 2012; Zamani et al., 2015). Sandin and Peters found that environmental impacts associated with current textile recycling techniques are related to their material and energy intensities, dependence on fossil fuel resources for energy, and transportation distance for waste collection (Sandin and Peters, 2018). Novel textile recycling techniques being developed must mitigate these

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sources of environmental impacts. Recently, textile waste valorisation via biological recycling enabled by microorganisms is being developed with the expectation that it will be relatively cleaner than conventional textile recycling approaches (Li et al., 2018).

One such biological recycling method was developed by our research group previously where PET fibres are recovered from waste polyester-cotton blends and glucose syrup is also produced as a value-added by-product (Hu et al., 2018; Li et al., 2018). The bio-recycled PET fibres recovered from this method can be reused as a renewable feedstock in the next textile production cycle. The biodegradable part of the mixed textile waste is subjected to alkaline pre-treatment and hydrolysis to convert it to glucose, while the remaining non-biodegradable part is processed into polyester and then re-spun into fibres. Recovery of PET fibre and glucose as by-product effectively from waste, also align with Circular Economy (CE) principles of converting waste to resource. Previously, blended polyester-cotton fabric waste was recycled biologically to produce (1) ethanol and polyester using alkali pre-treatment (Gholamzad et al., 2014) and (2) polyester and microcrystalline cellulose using phosphotungstic acid (Ling et al., 2019). While these bio-recycling techniques are expected to result in environmental benefits, there is a need to conduct a systematic assessment of environmental impacts associated with these emerging approaches. It is imperative to evaluate and improve the life cycle environmental impacts associated with such emerging technologies to ensure no unintended consequences, for instance, due to the use of a specific-chemical for treating textile waste, when they are eventually scaled up (Chopra et al., 2019).

The use of LCA is recommended for identifying hotspots in the textile supply chain (Nieminen et al., 2006). LCA studies conducted so far have presented the environmental life cycle impacts throughout the textile supply chain, and especially in relation to recycling techniques. However, none have quantified the environmental impacts of a textile bio-recycling method. Therefore, this knowledge gap in the literature will be addressed in this study. This paper will produce a life cycle inventory for a promising textile bio-recycling technology and identify unit processes that are hotspots for environmental impacts. There is a need to evaluate such novel bio-recycling methods from a life cycle perspective, identify the processes that contribute significant impacts, and guide future process efficiency improvements to reduce the life cycle environmental impacts. Findings from this study may inform whether the bio-conversion of textile waste to resources (PET and glucose, as a by-product) presents a potentially beneficial CE solution.

Against this background, this study aims to adopt LCA methodology to analyse the novel textile bio-recycling method, identify hotspots, provide recommendations to reduce the associated environmental impacts, and optimise the unit processes for further development. One kg of PET fibre recovered from 50–50% cotton-polyester blended textile waste fabric was chosen as the functional unit (FU). Energy consumption was quantified using Cumulative Energy Demand (CED), while midpoint and endpoint-based impact categories were computed through ReCiPe method. Results will inform researchers working on bio-recycling methods regarding the potential environmental impacts and avenues to optimise the involved unit processes before scale up to avoid unintended consequences.

1.1. Literature review

1.1.1. LCA in textile reuse and recycling

LCA is well established within the textile industry and has been used in the past to evaluate the environmental implications of different aspects of textile supply chains (Dahlbo et al., 2017; Kookos et al., 2019; Sandin and Peters, 2018; Van Der Velden et al., 2014; Youhanan, 2013; Zamani, 2011; Zhang et al., 2018). The life cycle perspective is widely considered essential for quantitative accounting of environmental impacts associated with textile recycling (Sandin and Peters, 2018). During our survey of the literature, we found that LCA has been applied in more than 40 textile recycling studies, and polyester is the most studied material (Hole and Hole, 2019; Ingrao et al., 2014; Landi et al., 2018; Leal Filho et al., 2019; Muthu et al., 2012; Norup et al., 2019; Sandin and Peters, 2018; Shen et al., 2011, 2010; Youhanan, 2013; Zamani et al., 2016). More details are available in the supplementary information (SI). Most of these LCA studies highlighted the environmental benefits of reuse and recycling over landfilling and incineration. Reuse and recycling of clothes are expected to induce a decline in virgin textile production (Sandin and Peters, 2018). However, considering the rapid reduction in the apparel life span due to fast-changing fashion trends (Januário et al., 2019), it is unlikely that reuse alone can manage the rapid increase in textile waste generation. While reuse of clothing is an important mechanism to delay the disposal of textile waste to landfills, the development of recycling technologies to recover new textile material is critical for establishing closed-loop textile processing at scale.

1.1.2. Textile recycling routes

Textile recycling methods are classified as mechanical, semi-mechanical, thermochemical, hydrothermal and chemical (Sandin and Peters, 2018; Shen et al., 2010). Mechanical recycling is currently being carried out in two ways: (1) direct extrusion of fibres from PET flakes and (2) pelletization of PET flakes followed by melt-spinning to produce fibres. A small amount of ethylene glycol is added to the melt-spun fibres to improve their final quality in mechanical recycling (Shen et al., 2010). In conventional thermal recycling, combustion of solid textile waste takes place at a very high temperature to generate heat energy. The thermochemical recycling process, like pyrolysis, was applied to address issues associated with oxygen inadequacy in thermal recycling processes as it leads to the generation of greenhouse gas (GHG) like carbon monoxide. Pyrolysis is a decomposition process in which nitrogen is the carrier gas instead of oxygen to create the inert atmosphere needed for combustion and also produces various products like activated carbon fibre, char, syngas, bio-oil (Piribauer and Bartl, 2019).

Chemical treatment is considered a promising textile recycling route to recover cotton fibres by dissolving the polymeric part (Yousef et al., 2019). In this chemical technology, cotton is recovered from textile waste using commercially available chemicals. It includes three main steps: (1) pre-treatment of the textile waste by leaching the textile dye using nitric acid; (2) dissolve all the organic materials in the pretreated waste fabric including polyester, using Dimethyl Sulfoxide (DMSO); (3) purify the finally recovered cotton fibre by bleaching it with sodium hypochlorite and diluted hydrochloric acid. The chemicals used were regenerated and can be reused. Environmental assessment of preliminary lab-scale experiments revealed that this technology had reduced GHG emissions calculated based on energy consumption, and the amount of materials (cotton and polyester) recovered (Yousef et al., 2019). However, the environmental impact assessment of other novel recycling approaches, such as biological recycling methods, are still under investigation.

1.1.3. Allocation procedures, impact categories, and data

The main methodological assumption in most of the LCA studies of textile recycling approaches in the literature is that waste sent for recycling is considered burden free from environmental impacts (cut-off approach), and the textiles made from recycled materials replace textiles made of virgin fibres (system expansion). A 1:1 replacement ratio is assumed, which may not be representative of the actual scenario. Also, climate change and energy use are the most studied impact categories, resulting in a risk of missing out on other relevant impact categories leading to burden-shifting and sub-optimisation (Sandin and Peters 2018). Most textile recycling systems modelled have used secondary data from Ecoinvent and Gabi databases, while few have used primary data. The primary data mainly includes energy data of the waste management facility and the recycling process (Sandin and Peters, 2018). Finding life cycle inventory data, especially for the end-

of-life of textiles, is a challenge considering the confidential nature of textile disposition. It is imperative to collaborate with technologists engaged in the research and development of emerging textile-recycling technologies for primary data to estimate their prospective environmental impacts (Chopra et al., 2019). For this reason, this study was conducted in close coordination with the researchers that developed the technology. While no LCA studies are examining bio-recycling, it is critical to evaluate the technology at the early stages of its development (Maclean et al., 2020).

The outcomes of this assessment can be beneficial to various stakeholders in the textile industry, government as well as researchers involved in evaluating environmental impacts of such biological methods, mainly due to primary data utilised and the detailed set of indicators assessed.

2. Methods

For the present evaluation, Environmental Life Cycle Assessment (ELCA), also commonly referred to as Life Cycle Assessment (LCA), was used. This methodology aims to measure the potential environmental impacts of a product/process throughout its life cycle encompassing different stages, from raw material extraction to production, distribution, use and disposal (ISO, 2006a). In this work, LCA is used to assess the potential environmental impacts of a textile bio-recycling method developed to recover PET fibre from textile waste fabric.

According to ISO 2006, in this study, LCA was conducted in four main phases: 1. Goal and Scope, 2. Life Cycle Inventory (LCI), 3. Life Cycle Impact Assessment (LCIA) and 4. Interpretation. A detailed description of the four phases can be found in SI. All the compiled data were processed using SimaPro 8.5 software, using the Ecoinvent version 3 database (www.ecoinvent.org) and ReCiPe 2016 (H) method for the impact assessment phase. The heirarchist version of ReCiPe, a commonly used consensus model, is adopted. Additionally, CED, a single issue-based impact assessment method, was used to calculate and compare the energy intensity of involved unit processes.

The inventory results were first translated into 18 midpoint impact categories such as climate change, ozone depletion, terrestrial acidification, indicating the negative impacts on the environment. The midpoint results were further grouped into three major endpoint categories: ecosystem, human health, and resource depletion, indicating the overall damage suffered. The midpoint approach enabled comparative assessment of production processes for each impact category in terms of the characterization values expressed as indicator units such as kg CO_2 eq. for Global Warming Potential (GWP). In the endpoint approach, the overall results are represented quantitatively as points or percentages. Processes, materials, and energy used, impact and damage categories can all be compared and analysed based on the points obtained in the endpoint method. This will enable decision-makers to identify critical points or issues in the studied process that are environmentally impactful and require improvements. The four stages of LCA in relation to the system studied in this work are described in sections below.

2.1. Goal and scope definition

This study explored the application of LCA to quantify the environmental impacts of the biological textile recycling method to recover PET fibres for sustainable end-of-life management in the clothing industry. The main goal of this LCA is to identify the environmental hotspots, including inventory item, sub-process, and impact category, that contribute a significant share of the impacts in the bio-recycling process for PET recovery. The biological method includes different steps from crushing to pre-treatment, enzymatic hydrolysis, and finally re-spinning to recover PET. The hydrolysate purification process used to produce glucose-rich syrup, a by-product from the bio-recycling method that can be used for other industrial applications, was also evaluated for its environmental impacts. Additionally, a scenario analysis was conducted to understand the difference in the potential environmental implications created by the reduction in the share of PET bottle chips added during the melt-spinning process.

In this study, *1 kg of recovered PET fibres* was chosen as the FU. All the relevant inputs and output flows (materials and energy) refer to this FU. *Gate-to-cradle* LCA was carried out, which means that life cycle impacts starting from the point of textile waste generation (gate) to conversion into a recovered PET fibre for a new textile application (cradle) are accounted. Since the 50–50 cotton polyester blended textile waste used as an input for the laboratory-scale bio-recycling process modelled in this paper comes from the industry partner, Hennes & Mauritz (H&M), a gate-to-cradle system boundary is considered. If the studied bio-recycling technology is scaled up such that the textile waste comes from the consumer at the end of life, then a grave-to-cradle study would have been more suitable. All processes in the bio-recycling method until the production of the final product, i.e., recycled PET fibres were included. The actual end use of the recovered PET, i.e., garment production, usages of clothes, washing, and disposal by consumers at their end of life were not considered to be within the scope of this study. Defining the cradle (i.e., raw materials) stage for recycled products is generally considered a problem, i.e., whether to include the first life cycle of the product or not. In this study, the conventional cut-off approach was used. The first life of the waste fabric that comes as the input for the bio-recycling was not considered. The fabric that enters the recycling process is considered waste, and hence a burden-free approach was employed. The detailed system boundary and process flow of the textile bio-recycling method used to recover PET fibre is presented in Fig. 1.

2.1.1. System description

2.1.1.1. Bio-recycling method. The overall process flow for the bio-recycling method of a 50% cotton and 50% polyester waste fabric used to recover polyester and glucose analysed in this study is depicted in Fig. 2. A detailed description of the process flow can be found in Li et al., 2018, while a brief description is provided below.

2.1.1.2. Crushing. The macrostructure of fibres decreases the efficiency of biological catalysts (enzymes) in enzymatic hydrolysis due to the lack of surface area required to degrade the natural components of the textile. For this reason, the textile waste fabric was first crushed into small pieces by a double shaft shredder to increase the surface area for the pre-treatment process (Li et al., 2018).

2.1.1.3. Pre-treatment. For enhancement of the susceptibility of the waste to hydrolysis with a reduction of the crystallinity of the cellulosic fibres, a promising pre-treatment method was constructed with reusable chemicals. The ground textile was mixed at a solid loading of 5 w/v% with a solution of 7 w/v% NaOH and 12 w/v% urea at −20 °C in a freezer for 6 h. Then it was neutralized by the addition of HCl acid and washed with tap water (Li et al., 2018). The textile waste sample was dried to a constant weight in an oven at 40 °C for 48 h (Hu et al., 2018).

2.1.1.4. Enzymatic hydrolysis. This process hydrolyzes the cellulose from cotton into soluble glucose, while the non-biodegradable material (e.g., polyester) remains intact and is separated as a solid through filtration. Glucose recovery from the treated textile waste depends on the proportion of natural fibre in its composition. The pretreated textile waste was blended with the enzyme solution (Novozymes®) and went through hydrolysis in a bioreactor. The pretreated textile sample was added into sodium citrate buffer (50 mM, pH 5) at a solid loading of 3 w/v%. In a bioreactor, 20 FPU/g commercial cellulase and 10 U/g commercial β-glucosidase were added to the textile solution at 50 °C and pH 5.

2.1.1.5. PET recovery/purification of hydrolysate. PET residue from the

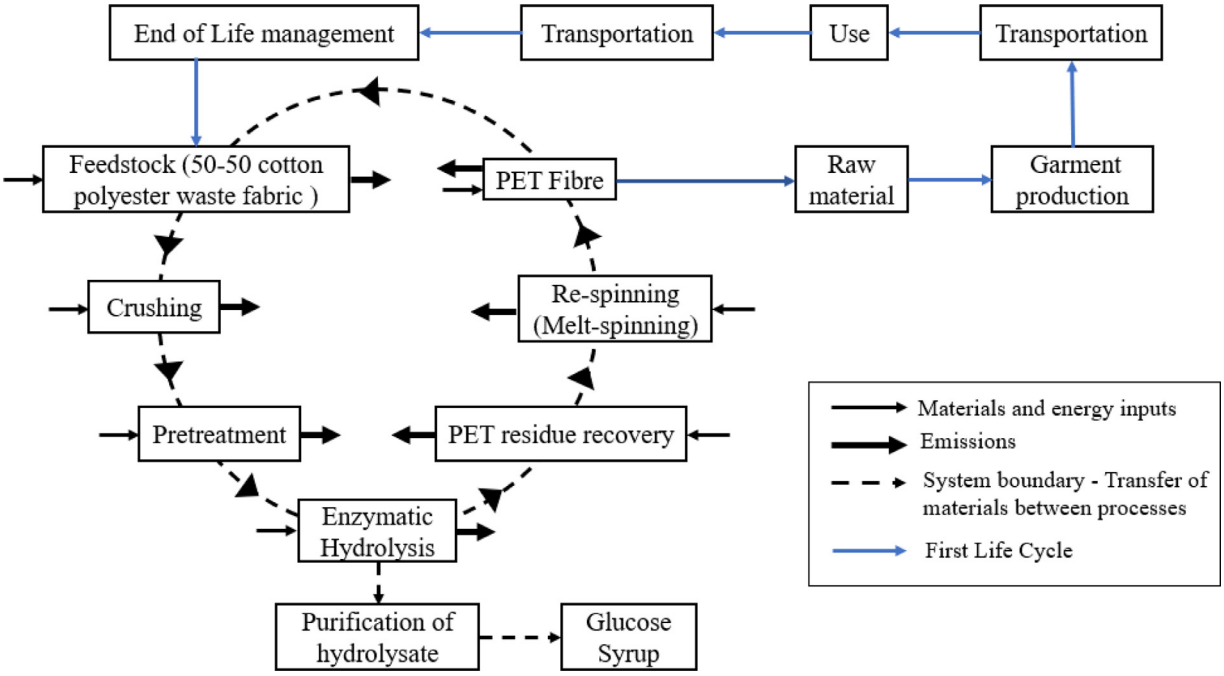


Fig. 1. Illustration of the system boundary.
Note: System boundaries description: Cotton-polyester textile waste blends at its end of life are biologically processed (pre-treatment, hydrolysis, purification, and melt-spinning) and PET fibre is recovered as output. The recovered PET fibre becomes the feedstock for a new garment production cycle represented by blue arrows.

glucose-rich hydrolysate solution was recovered by filtration before the spinning step. The hydrolysate was converted into a glucose-rich syrup for further usage in different bio-based products such as bio-plastics, bio-surfactants (cleaning agents), and bio-chemicals in industries. The hydrolysate was purified from the chemicals by the addition of activated carbon at a loading rate of 15 w/w%. After the removal of activated carbon by paper filtration, the hydrolysate passed through ion exchange columns for the removal of ions. Later, the columns were cleaned with deionized water and were regenerated by the addition of H2SO4 (50 w/w%) and NaOH (7 w/v%) to the resins. Washing of the

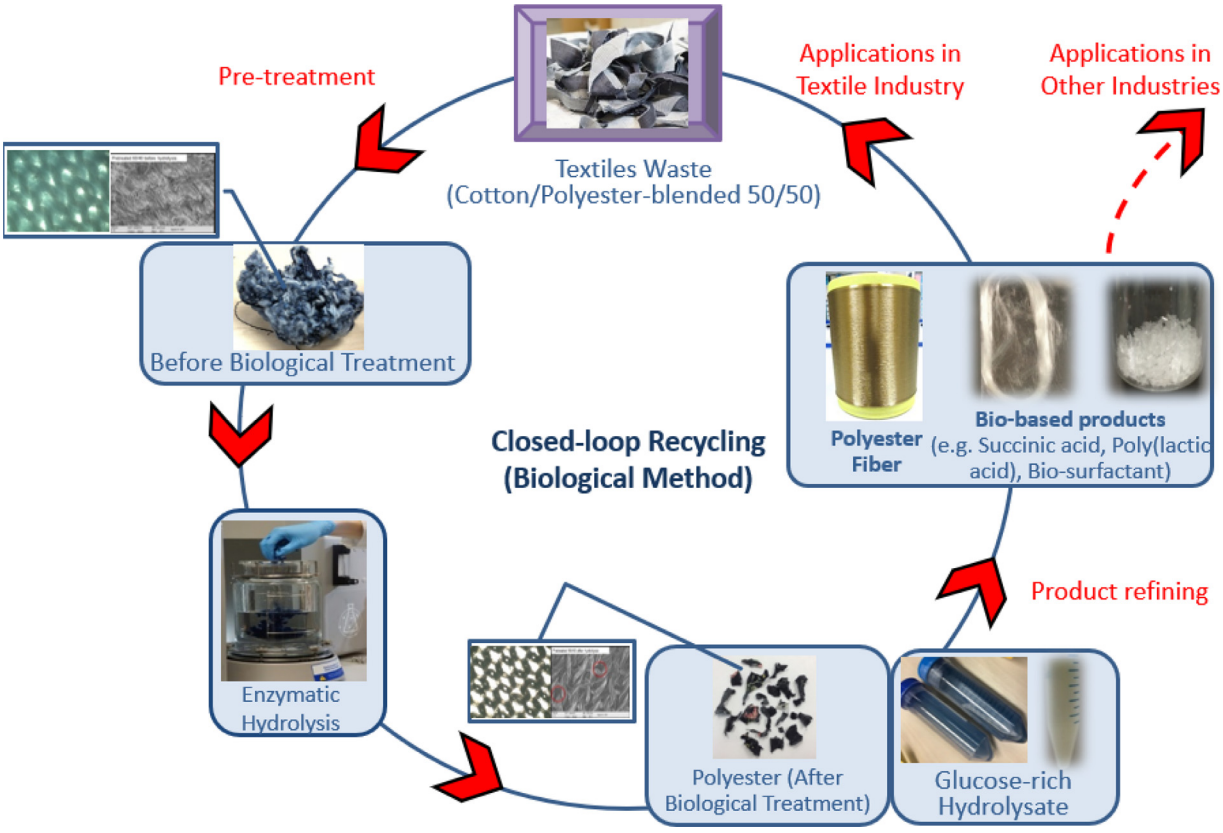


Fig. 2. Process flow of the bio-recycling method for converting 50–50 cotton/polyester waste fabric to PET fibre and glucose syrup.

Table 1
Summary of LCI data to produce 1 kg of polyester through bio-recycling.

Process Specification	Materials (kg)	Amount	Electricity (kWh)	Water (Kg)	Steam (Kg)	Data Source
Crushing			0.54	–	–	Calculations based on our previous study (Li et al., 2018)
Pretreatment	NaOH	0.68	10.09	1.46	–	Stoichiometric calculations based on our previous study (Li et al., 2018) & ICR of the TEA; Electricity, water and steam: MSR of the TEA (Kwan, 2018)
	Urea	1.17				
	HCl	0.62				
Enzymatic Hydrolysis	NaOH	0.10	0.08	16.2	3.54	Stoichiometric calculations based on our previous study (Li et al., 2018) & ICR of the TEA; Electricity, water and steam: MSR of the TEA (Kwan, 2018)
	Citric acid	0.16				
	Cellulase	0.14				ICR of the TEA (Kwan, 2018)
	Beta glucosidase	0.014				
PET Recovery/ Purification of Hydrolysate	Activated carbon	0.0002	0.0005	0.16		Stoichiometric calculations based on our previous study (Li et al., 2018) & ICR of the TEA; Electricity, water and steam: MSR of the TEA (Kwan, 2018)
	H_2SO_4	0.0041				
	NaOH	0.0016				
	Monosphere 88 Resin (Cationic)	0.1154				
	Monosphere 66 Resin (Anionic)	0.1154				
Re-spinning	PET Bottle Chips –	0.80	0.10	0.84	7.34	ICR of the TEA (Kwan, 2018)

remaining hydrolysate from the columns decreased the glucose recovery. Lastly, the hydrolysate was evaporated to obtain the glucose syrup.

Re-spinning (Melt-spinning) – For further applications in the textile industry, the recovered polyester from hydrolysis was re-spun into PET fibres. A suitable tenacity of fibre is needed for weaving or knitting processes., which was achieved in three steps: granulation, solid-state polymerization, and melt-spinning. First, the recovered PET residue from the hydrolysis was washed, dried, and sent to the granulation step for pellet formation. Then the polyester pellet was obtained after cutting. The second step, Solid State Polymerization (SSP), was conducted to increase the molecular weight of recovered polyester pellets. Recovered polyester pellets were collected after heating at 230 °C for 1800 min. Finally, PET bottle chips were added to the recovered polyester pellets in the melt-spinning process in a ratio of 80–20 to obtain PET fibres suitable for textile application.

2.2. Life cycle inventory

The LCI phase deals with systematic data collections regarding the studied process. The material and energy flow for each step in the textile bio-recycling method is quantified exhaustively. A detailed description of the inventory collection and analysis for the bio-recycling method is presented in Section 2.2.1.

2.2.1. Bio-recycling method

All the relevant and available data regarding the bio-recycling method were collected and converted in accordance with the FU considered. The LCI for this study was based on primary data extracted using the previous experimental research conducted by our research team (Li et al., 2018). Data related to the processes that represent materials and energy flow were chosen from the Ecoinvent. All the inventory results were allocated based on the mass of the product. The geographical boundary includes Hong Kong and Taiwan, depending on the process flow.

Energy requirements (electricity, heating, cooling, freezing, drying) for crushing, pre-treatment, enzymatic hydrolysis, hydrolysate purification, and melt re-spinning were normalized as per the FU. This data was obtained from the Itemized Cost Report (ICR) and Materials & Streams Report (MSR) of the Techno-Economic Analysis (TEA) carried out by our collaborators previously (Kwan, 2018). The report was generated based on the pilot-scale development of the textile bio-recycling technique for the recovery of PET and glucose by processing 10 Metric Ton (MT) or 10,000 kg of textile waste per hour. The machine specifications, power, and capacity of the equipment, the number of hours needed to process the textile waste and other materials were obtained from another investigation conducted by our research team (Li et al., 2018). The data for electricity mix, heat and power generation processes, water, and steam were obtained from the Ecoinvent database.

The mass of the materials and chemicals, such as sodium hydroxide (NaOH), urea, citric acid, hydrochloric acid (HCl), used in the pre-treatment and enzymatic hydrolysis step was calculated based on the solid-to-liquid ratios and weight-to-volume ratios of the solutions, which was determined by stoichiometry based on molar mass. HCl was added to the textile waste to neutralize the alkalinity in the final step of pre-treatment. The textile waste may contain a variety of chemicals, so to generalize this step for all types of textile waste fabrics, NaOH was assumed to be the only source of alkalinity in the textile waste (Li et al., 2018). Therefore, the mass of HCl needed for pre-treatment was calculated based on the amount needed to neutralize the NaOH. The data for chemicals' production were retrieved from the Ecoinvent database as well.

In the enzymatic hydrolysis step, the LCI data for the production of cellulase enzyme was based on the process used by the Novozyme company. Data for β-glucosidase was not found in Ecoinvent. Moreover,

the LCI for β-glucosidase was neither available from existing literature sources nor commercial databases. Degradation of the highly crystalline structure of cellulose requires the synergy of endoglucanases, exoglycanases, and β-glucosidases in a complete cellulase system (Hu et al., 2018). Commercially, β-glucosidases are used along with cellulase only if the cellulase enzyme does not contain enough β-glucosidases needed for the hydrolysis. In this work, it was assumed that the cellulase enzyme used contains enough β-glucosidase for hydrolysis. Hence, only the LCI data for cellulase enzyme was used for modelling the hydrolysis step (Table 1).

The material composition of the textile waste determines the amount of glucose extracted during enzymatic hydrolysis. Likewise, the load of hydrolysate determines the amount of activated carbon needed to purify the glucose-rich hydrolysate solution. The amount of activated carbon used in the hydrolysate purification step was directly extracted from the TEA (Kwan, 2018). The processes related to the production of activated carbon, sulfuric acid, cationic, and anionic resins used in the hydrolysate purification process were all collected from Ecoinvent.

For the melt re-spinning process, the polyester recovered from textile waste, through enzymatic hydrolysis was sent to Taiwan. PET bottle chips (80% w/w) were added to the total weight of the recovered polyester in order to produce recycled PET fibres that are functionally equivalent to virgin PET fibres. According to the FU of this work, 80–20 is the ratio of PET bottle chips used to recover polyester chips. Simply put, to produce 1 kg of recycled PET fibres, we need 80% PET bottle chips and 20% of recovered polyester from 50:50 cotton-polyester waste. The PET fibre production process includes pellet production, melt-spinning, and drawing processes. The LCI data for electricity, water, and steam needed for re-spinning were all extracted from the ICR of the TEA, in accordance with the FU. The production processes related to waste PET bottles and the energy consumption were chosen from Ecoinvent.

Table 1 presents the summary of LCI data used in the bio-recycling method and its sources. Details of all the stoichiometric calculations, including but not limited to the ones mentioned above, are given in the SI. In LCA, results derived by using primary data are generally considered more reliable than secondary data sources. Hence, in this study, the LCI is created using calculations and estimations from the actual bio-recycling carried out in a pilot plant. LCAs of emerging technologies can provide experimentalists with results that reflect the environmental burdens of each unit process, which can be reasonable estimates for process optimization (Chopra et al., 2019). Such emerging technologies, when optimized based on LCA results, may produce lesser environmental impacts when scaled-up. However, it is unlikely that the LCA results of an industrial-scale implementation of this emerging technology would be comparable as they depend on many factors like the region, the source of energy used, amongst many others. A sensitivity analysis is included here to ensure the reliability of the LCA results.

3. Results and discussion

3.1. Life cycle impact assessment (LCIA) and interpretation

3.1.1. Midpoint and endpoint assessment results for the textile bio-recycling method

The performance of the bio-recycling method used to recover 1 kg of PET fibres from 50–50 cotton-polyester blended textile waste fabric, in terms of midpoint-based environmental impact categories and three endpoints are described below. Midpoint results in Fig. 3 illustrate the percentage contribution of each step in bio-recycling to the impact categories. The results are presented in four main categories: feedstock, pre-treatment, process, and product.

Fig. 4.

The feedstock of 50–50 cotton-polyester blended fabric used in bio-recycling, is a waste considered to be free of environmental burden. In other words, a cut-off allocation procedure is employed for the material inputs. Pre-treatment contributes the most (around 60%) to majority of the midpoint categories except land use (38%), marine eutrophication (28%) and ozone depletion (14%) when compared to the other steps in bio-recycling. These impacts are caused by the process energy and chemicals used for the alkali treatment. Specifically, the major contributors are production of urea, cooling, neutralization, and heating. Process energy-related impacts can be further traced back to the hard coal mine operation in electricity production. Hard coal mine operation and ammonia production are also significant contributors to global warming-related impact categories. Ammonia production dominates ecotoxicity-related impact categories. Urea production contributes the most to water consumption-related impacts. Hard coal mine operation and natural gas production are also significant contributors to the fossil resource scarcity impact category. Within pre-treatment, crushing was found to be the least impactful unit process. amongst the processes, melt-spinning and enzymatic hydrolysis are the most environmentally detrimental. Melt-spinning mainly affects fossil resource scarcity (27%), terrestrial ecotoxicity (29%), and mineral resource scarcity (21%). PET bottle chips (80% w/w) are added to the biologically recovered polyester from the 50:50 waste blend, and then subjected to melt-spinning in order to improve its final quality (we revert to this aspect in Section 3.3.). Fossil fuel scarcity impacts originate from xylene production, a subprocess in the PET granulate, used in the production of PET bottles. Heat production for the industrial furnace to generate steam, one of the process energies used, contributes the most towards terrestrial ecotoxicity. Iron mine and copper mine operation within the electricity production processes used as process energy for PET bottle production contributes the most to mineral resource scarcity.

Enzymatic hydrolysis causes the highest impact on marine eutrophication (64%), and land use (49%). Citric acid produced in China using the submerged fermentation process contributes the most to marine eutrophication and land use. The process energy, including electricity and steam, and other chemicals, like NaOH and cellulase enzyme, contribute less to the remaining impact categories. Pellet extrusion and hydrolysate purification were found to be less environmentally detrimental when compared to the other processes within bio-recycling. Pellet extrusion contributed the most (around 12%) to most impact categories except water use, human toxicity and ozone depletion. The impacts can be traced back to the process energy used, which includes petroleum and gas refinery operations within electricity production. Hydrolysate purification was found to be least environmentally impactful in terms of most impact categories (less than 10%) except stratospheric ozone depletion. The impacts can be traced back to the trichloromethane production process, which is used in ion exchange columns in hydrolysate purification.

Like midpoint results, pre-treatment caused the highest impact on all three endpoints: human health, ecosystems, and resource depletion, compared to the other steps in bio-recycling. Electricity production, hard coal mine operation, and ammonia within urea production are the main contributors to ecosystem system quality (59%) and human health impacts (61%). Hard coal mine operation and natural gas production within urea production, contribute the most to resource depletion. Production of NaOH and HCl used in pre-treatment cause much lower impact.

Melt-spinning is the next most impactful step, and it mainly affects the resource depletion category (34%) and is comparatively less impactful on ecosystems (14%) and human health (15%). Process energy used in steam production as well as xylene used in PET bottle production, are responsible for the impacts on mineral resource and fossil resource scarcity. Enzymatic hydrolysis is the next most dominant step, and it mostly affects ecosystem quality (14%) and human health (12%). This step causes comparatively lower impacts on mineral resource and fossil resource scarcity and hence consequently is less detrimental to resource depletion (8%). Like the midpoint results, pellet production, hydrolysate purification, and crushing were found to be less harmful in

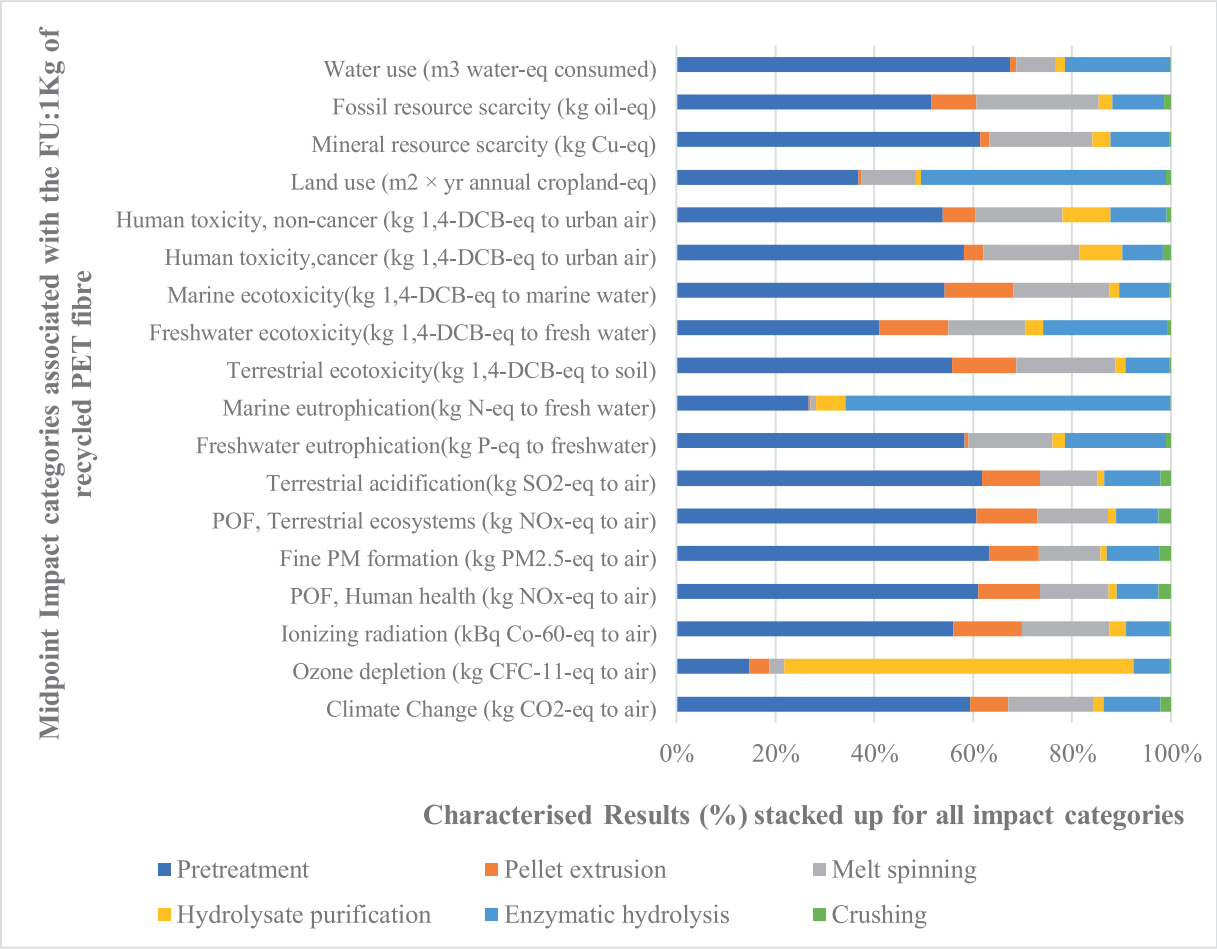


Fig. 3. Midpoint assessment results of the Textile Bio-recycling method used to recover 1 kg of PET fibres using ReCiPe (H) impact assessment method (Note: POF-Photochemical Oxidant Formation. The results are stacked up for all unit processes. Each coloured bar represents the% contribution of a particular process on that impact category).

terms of environmental impacts. amongst the three endpoints, all steps excluding pre-treatment contributed more towards to resource depletion when compared to ecosystem quality and human health.

3.1.2. Cumulative energy demand for the textile bio-recycling method

Midpoint and endpoint evaluation indicate that the bio-recycling method is sensitive to the process energy used. Hence energy use of the studied system was further evaluated using the CED method. Fig. 5 presents the CED results for 1 kg of recycled PET fibres produced by the bio-recycling method. The energy use is represented by non-renewable sources (fossil, nuclear, biomass) and renewable sources (biomass, wind, solar, geothermal, water). The most energy-intensive steps are pre-treatment (207 MJ) followed by melt-spinning (98.5 MJ) and enzymatic hydrolysis (44.8 MJ). Crushing (5 MJ), pellet extrusion (32 MJ), and hydrolysate purification (11 MJ) contribute comparatively less to the overall energy demand. Pre-treatment (freezing and drying) significantly contributes (87%) to the Non-Renewable Energy use (NREU), mainly through the use of chemicals (natural gas used (20 MJ) in the production of ammonia liquid within urea production) and electricity (hard coal mine operation (87 MJ). The other significant

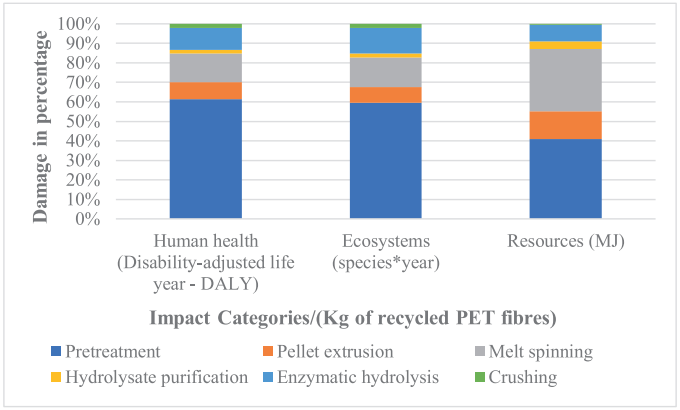


Fig. 4. Endpoint results for bio-recycling method, per kg of PET recycled fibre.

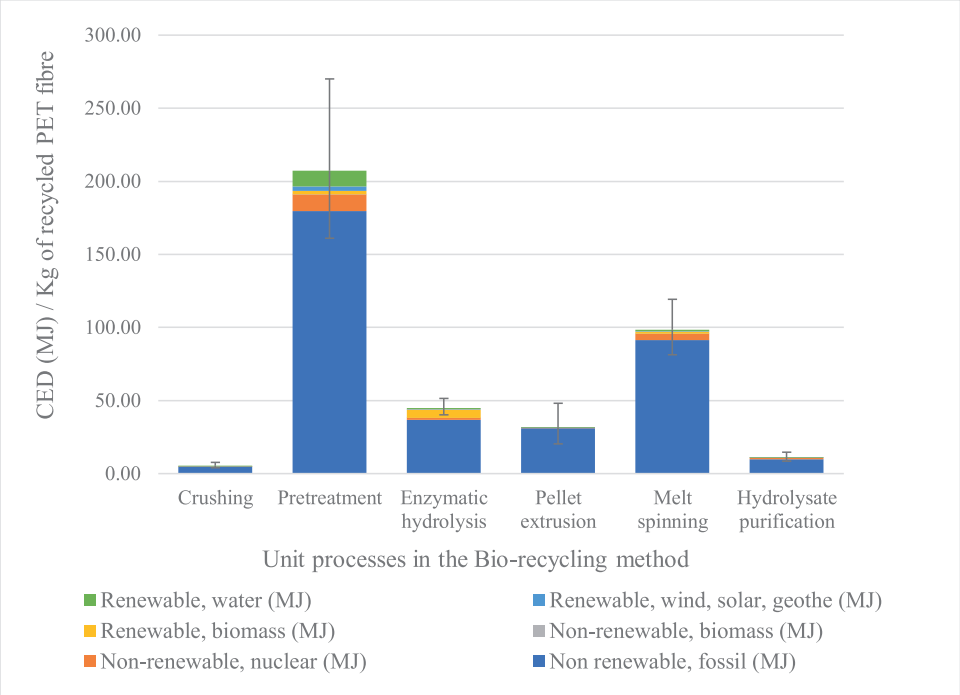


Fig. 5. LCA results for Cumulative Energy Demand based on Hong Kong and Taiwan's Electricity Mix for the unit processes in the bio-recycling method. Error bars reflect the 2.5th and 97.5th percentiles, constructed via Monte Carlo Simulation to account for uncertainty. (Note: The standard data uncertainties included in Ecoinvent were only considered here).

contributor to NREU is melt-spinning through the use of steam and energy consumption in the production of xylene, ethylene glycol, and other subprocesses in PET bottle production.

The Hong Kong electricity mix was used for modelling all the steps of bio-recycling except melt-spinning. Melt-spinning was modelled using Taiwan electricity mix, as it was carried out by the Industrial Technology Research Institute (ITRA) in Taiwan. Hong Kong electricity mix in Eco invent represents the electricity imported from China southern power grid. Coal is the primary fuel source (52%) for electricity, followed by natural gas. For Taiwan, oil is the primary fuel source for electricity. Both (Hong Kong and Taiwan electricity mix) represent electricity from non-renewable resources. Pre-treatment and melt-spinning included more energy-intensive sub-processes and chemicals, when compared to other steps in bio-recycling. As a consequence, production of chemicals and electricity usage contributed more to NREU. However, LCA results might vary for bio-recycling carried out in other places based on the countries and energy resources used to calculate impacts.

3.2. Sensitivity analysis for bio-recycling

Tornado diagram is used to present the sensitivity analysis results for different input parameters. For this study, sensitivity was performed to understand to what extent an output parameter (i.e. GWP of producing 1 kg of recycled PET fibre) changes in response to +/- 10% change in input parameters. The most impactful input parameters within each unit processes were identified and varied systematically, one at a time. Model parameters that were considered in this analysis include process energy and urea used in the pre-treatment step, PET bottle chips in the melt spinning step, and citric acid and cellulase enzyme used in the enzymatic hydrolysis step of the biorecycling method. Fig. 6. highlights the relation between the five most impactful model inputs and the overall GWP results. We note that the GWP results are relatively more sensitive to the process energy and urea consumption during the pre-treatment step. Therefore, additional research could improve the environmental performance of pre-treatment inputs, including: a) energy efficient processes (freezing and drying); b) alternative chemicals with improved environmental performance. Prioritizing research questions based on the sensitivity results may fast-track reductions in environmental performance for textile bio-recycling. This may, in turn, justify greater research efforts and investments to examine scalability of this emerging technology in real-world conditions.

3.3. Scenario analysis: addition of pet bottles to recovered polyester ratios in melt- spinning

PET bottles are added to the recovered polyester as a final step in the melt-spinning process to obtain industrial-grade PET fibres suitable for textile application. The impact of the amount of PET bottle chips on the production of PET fibres is analysed by developing different scenarios. The baseline 80–20 scenario analysed in this work refers to the addition of 80% PET bottle chips to the polyester recovered from a blend of 50–50 textile waste to produce 1 kg of PET fibres. Four additional scenarios of PET bottles to recovered polyester percentage ratios (60–40; 50–50; 40–60; 20–80) were compared with the baseline scenario to understand the environmental implications of reducing the percentage of PET bottles added to the process. The ReCiPe method was used to calculate the impacts for all three endpoints, and the baseline scenario of 80–20% resulted in the least harmful (Fig. 7). It is noteworthy that as the ratio of PET bottle to polyester decreased in the scenarios, the impacts on all the three endpoints increased. As the percentage of the recovered polyester (output of the enzymatic hydrolysis) used in the melt-spinning step was increased, the process energy and materials used in the biological method rose. Additionally, the spinning of high quality fibres from recovered PET residue is more resource intensive in comparison to spinning fibres from PET bottle chips. It is important to note that decreasing PET bottle ratios in the melt-spinning step alone cannot ensure reduction in the environmental impacts. The energy and material requirements, and the quality of the polyester pellets recovered from the bio-recycling process influence the environmental performance significantly.

4. Conclusions, future work, and limitations

In this article, the environmental hotspots for a novel textile bio-recycling method were identified. It was found that the pre-treatment and melt-spinning were the two most environmentally detrimental steps across all the unit processes analysed within textile bio-recycling.

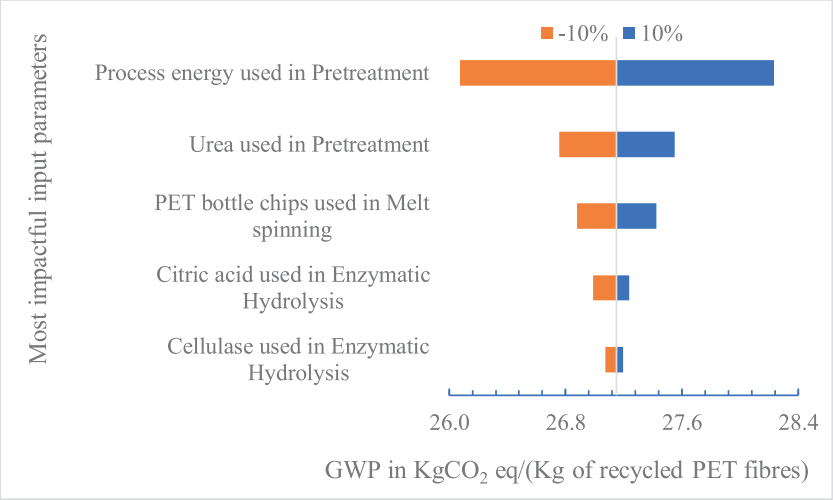


Fig. 6. Tornado plot to illustrate the impact of +/- 10% changes in each input parameter on the output value i.e. GWP of producing 1 kg of recycled PET fibre. Only the most impactful inputs within each unit process are shown. Blue bars indicate the GWP results for increases in input parameters. Orange bars indicate the GWP results for the decreases in input parameters. The axis line in the centre represents the baseline GWP results.

An analysis of the CED showed that pre-treatment and melt-spinning steps contribute to non-renewable energy usage, as these processes take place in Hong Kong and Taiwan, where the electricity grids are predominantly fossil-based. Moreover, the pre-treatment step generates substantially higher impacts compared to other steps because of energy-intensive subprocesses like cooling, heating, neutralization, and production of chemicals like urea. Tornado plots were used to present the sensitivity information. It was found that pre-treatment step was more sensitive towards process energy and chemicals used. Scenario analysis was conducted to understand the environmental implications of altering the PET bottle ratios. It was found that the reduction in PET bottle percentages alone cannot reduce the environmental impacts, quality of the recovered polyester pellets used to spin the fibres in the melt-spinning step is also crucial in determining the environmental impacts. Bulk of the impacts related to fossil fuel scarcity were related to the PET bottle production, the main component in the melt-spinning step for improving the fibre quality of the recovered polyester pellets. Hence, it can be concluded that the process energy and the materials and chemicals used in pre-treatment and melt-spinning steps have to be further optimized to reduce the environmental impacts associated with this emerging valorization technology.

Techno-Economic Analysis (TEA) of the bio-recycling method was conducted by our research team previously to determine the economic performance (Kwan, 2018). Results reflected an overall deficit contributed mainly by raw material cost (commercial cellulase and PET bottle chips). Similarly, our LCA study emphasized that PET bottle chips added in the melt-spinning step and cellulase used in the enzymatic hydrolysis step are responsible for considerable environmental impacts. From the scenario analysis presented in this study, it is also clear that the reduction in PET bottle chips alone cannot fully mitigate the environmental impacts. Currently, the reduction of PET bottle chips usage is not feasible, as it will affect the quality of the final product, i.e., recycled PET fibre, and it cannot be assumed to substitute virgin PET fibres functionally. Further investigations must focus on optimizing these economic and environmental hotspots to develop this technology in a sustainable manner.

Therefore, future research should focus on 1) improving the existing pre-treatment method used to facilitate enzymes to access the internal structure of the cellulose in the fibre, and 2) evaluating optimal conditions for enzymatic hydrolysis with minimal enzyme and energy inputs, to improve the final quality of the recovered polyester pellets used in melt-spinning. While it is not possible to predict if the improved unit processes will demonstrate better environmental performance, the sensitivity analysis showed that improvement in just one most impactful parameter (decrease in electricity consumption in the pre-treatment step) alone could decrease the overall GWP results of the bio-recycling method. Hence, a combination of several improved input parameters can considerably reduce the GWP results of the bio-recycling method. LCA can be used to validate this and enable a better understanding of whether this bio-recycling method can be scaled up. Improving the quality of the recovered polyester pellets from hydrolysis, will also consequently reduce or remove the amount of PET bottle

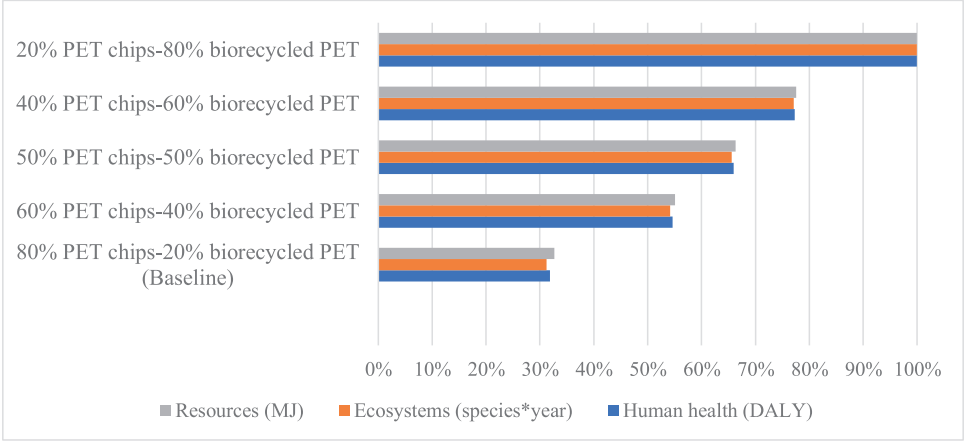


Fig. 7. Relative Environmental Impacts associated with the Five Scenarios with varying the ratio of PET bottle chips to Recovered Polyester pellets. Endpoint results from ReCiPe were normalized to the highest impact for ease of comparison.

chips commercially bought, resulting in a reduction of costs. TEA results also showed that glucose syrup and recycled PET fibres proved to be a good return on investment. The application of this technology to treat other textiles such as silk, nylon, and wool may also improve the economics of the bio-recycling method in the future (Kwan, 2018).

LCA mainly depends on scenarios, and the results obtained have to be interpreted from that perspective. Scenario uncertainty related to the choices made in terms of the allocation methods chosen, especially while modelling recycling systems, can lead to the variation in results. Conventionally (including this work), only the cut-off approach is used to analyse the environmental burdens of the recycling approach. In the future, allocation at point of substitution, in which recyclable materials come with a burden from their previous life can be applied, to check whether results vary based on the allocation method chosen. Also, the system expansion approach can be included while modelling the impacts. For example, glucose, a by-product produced in bio-recycling, is considered as an alternative to the glucose produced in markets. The total emissions associated with the bio-recycling method can be determined by subtracting the emissions associated with saved glucose (by-product) production from the emissions associated with the production of PET fibre and glucose. Moreove, like any emerging technology, this textile valorization approach may result in changes to the production and consumption patterns within the textile industry. The environmental benefits of bio-recycling may be offset by the impacts created by the changes in the production and consumption of upcycled textile waste in the apparel sector. Thus it may be interesting to include the rebound effect in future assessments (Sandin and Peters, 2018). Additionally, an analysis of conventionally recycled and bio-recycled PET fibre in their use phase would add further value to this research. It is valuable to find out whether a garment made out of biologically recycled PET fibres has an extended, similar or shorter use phase (life span) when compared to garments made out of conventionally recycled PET fibres. Lastly, future work may also consider social aspects represented by the human, organizational, and societal well-being of all the relevant stakeholders in the textile waste recycling industry to provide a holistic sustainability assessment.

Regarding the limitations, it is essential to note that LCA studies generally are associated with many uncertainties, and the results obtained can be considered valid only for the kind of assumptions made to model the life cycle of the studied system. Each life cycle stage in itself has some critical uncertainty sources. For instance, variation in electricity production data, geographical location, can all result in reduced accuracy of inventory and lead to uncertainties. Overall, data quality, methodological choices made, and allocation procedure followed can all affect the LCA results. Hence, we conclude that LCA based approach is an appropriate methodological platform to understand the environmental implications of a system at its developmental stage (lab scale) itself and guide optimization of unit processes for reduced environmental impacts while scaling it up (Chopra et al., 2019). However, all LCA results have uncertainties associated with them and must be considered to make informed decisions.

Credit author statement

Karpagam Subramanian: Methodology, Conceptualisation, Formal analysis, writing-original draft

Shauhrat S Chopra: Conceptualisation, Resources, Supervision, Project administration

Cakin Ezgi: Formal analysis, visualisation

Xiaotong Li: Investigation

Carol Sze Ki Lin: Supervision, Fund acquisition

Conflict of interest statement

We confirm that this manuscript has not been previously published and is not currently under consideration by any other journal. Each named author has contributed toward the underlying research and drafting this manuscript. **Moreover, the named authors have no conflict of interest to the best of our knowledge.**

If you need any additional information, please let me know. We thank you for your consideration of this manuscript and look forward to your decision.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.resconrec.2020.104989.

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